Kant and Modern Mathematic

When I was young, I read Kant's *Prolegomena* where he outlines his ideas for his main book *Critique of Pure Reason*. In his Prolegomena, he outlines the following "pure concepts of understanding":

Transcendental table of concepts of the understanding

1.
According to quantity
Unity (measure)
Plurality (magnitude)
Totality (the whole)

2.
According to quality
Reality
Negation
Limitation

3.
According to relation
Substance
Cause
Community

4.
According to modality
Possibility
Existence
Necessity

(a) Kant's pure concepts of understanding

Logical table of judgments

1.
According to quantity
Universal
Particular
Singular

2.
According to quality
Affirmative
Negative
Infinite

3.
According to relation
Categorical
Hypothetical
Disjunctive

4.
According to modality
Problematic
Assertoric
Apodictic

(b) Kant's logical table of judgments

Figure 1: Side-by-side figures of Kant's tables.

These ideas stuck when I started studying mathematics; I couldn't help but draw parallels to modern mathematical thinking to these concepts. In this article, I will flesh out my thoughts intersecting these fields.